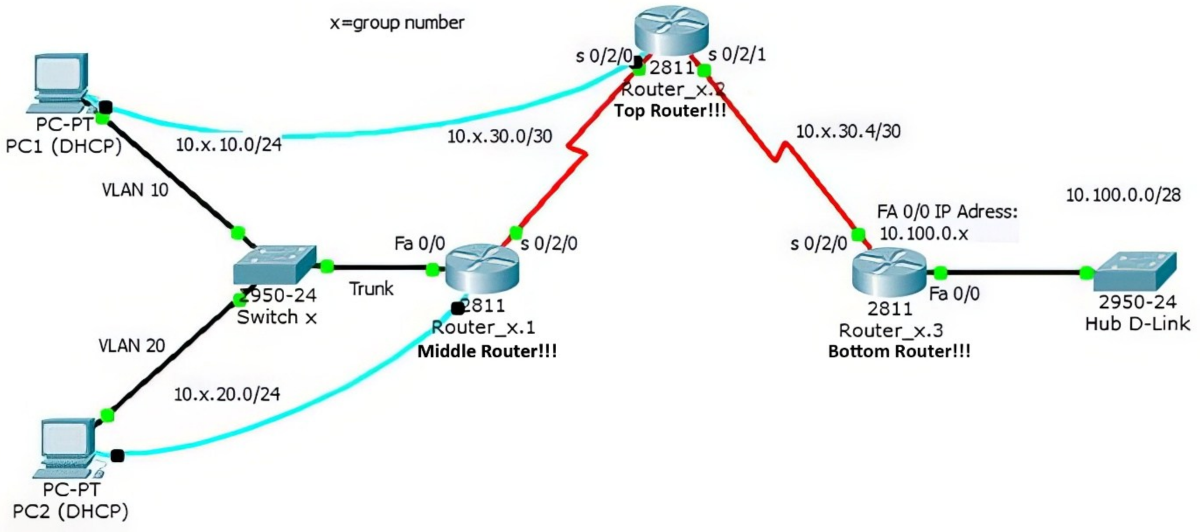
Challenge Lab: Create a network topology using real hardware



Preparations:

Before you start, you will be assigned a group number and a group of hardware components by the lab administrator. At the moment, the Hochschule Bremen networking lab can support up to 7 hardware groups. Each group will use an IP addressing scheme that is related to the group number. Using different IP address ranges will allow all groups to merge their topologies to an extended single topology later.

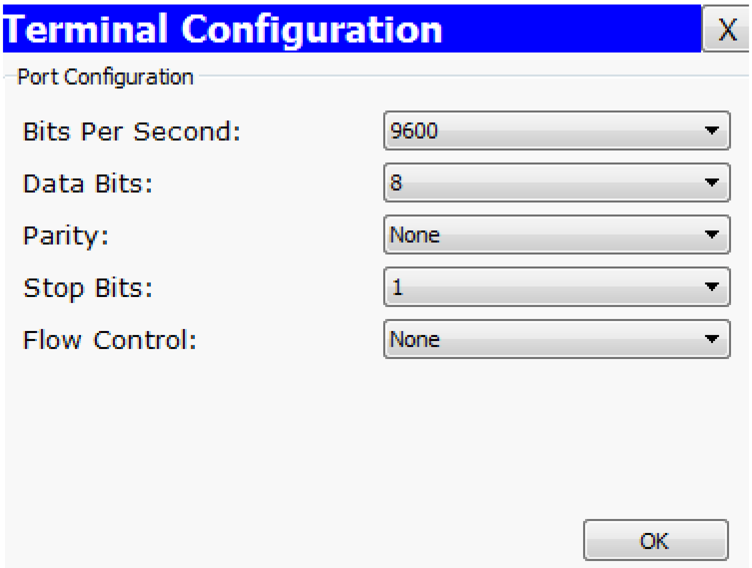
The serial cables are already attached to the routers. Please do not change the serial cablings.

Before starting, discuss the tasks within your group and assign the two lab PCs as either PC1 or PC2. Make sure to use the correct order of routers. The router with two serial connections of your hardware group should be designated as Router_x.2 and will later provide the clocking for both serial connections. Currently, this is the topmost router.

Prepare the labs PCs by accessing “VMWare Workstation” (use desktop shortcut) and start Windows10 as a virtual machine on each PC. When the VM is started, access the properties of the network connections. Each VM has an internet and a lab LAN connection. Only one of these connections can be active at the same time. To use the lab hardware, activate the lab LAN connection and disable the internet connection. Keep in mind that the lab LAN connection still may need IP address configuration after activating.

1. Cable your topology. The serial cables on the routers are already attached. For cabling the PCs, use the patch port information provided for every PC. Connect the Ethernet ports of the PCs to the Cisco Switch of your hardware group. Additionally connect the console ports of the PCs to the console ports of the routers. Unlike with packet tracer the routers need a physical connection to their console ports to access the CLI.

In order to access the CLI of the routers over the console port the PCs need to start a terminal program like Hyper Terminal or Tera Term. Start the terminal program and use the COM1 port of the PC for this connection.
Configure the terminal session with the following parameters:



Note: After the terminal session is correctly configured and the routers have booted successfully, press the <enter> key to access the prompt. Whenever there is no startup configuration present on the router you will be asked to use a setup dialog:
<Continue with configuration dialog? [yes/no]: >
To start with the manual configuration of the router, quit the configuration dialog with <n> or use the break command <control+c>.

2. Create the basic configuration for all 3 routers. Note that you will have to configure three routers but you only have two PCs with a terminal program for configuration. One PC will need to switch the console cable attached to one router after the initial configuration is complete. Configure the following for each router:
   - Hostname
   - Enable secret set to “class”
   - Console password set to “cisco”
   - The first 5 vty connections to “cisco”
   - IP addresses for all interfaces except FA0/0 on Router1. Router2 will be the DCE for both serial connections using a clock rate of 64000.

3. From the routers ping the neighboring interfaces. R1 should be able to ping R2. R2 should be able to ping R3. Why can R1 not ping R3?

4. Configure your switch using the console cables that are already attached to the switches. The configuration process of a Switch is similar to the confirmation of a router and also uses the same terminal connection to the console port. Configure the following:
   - Create VLANs 10 and 20.
   - Assign the switch port that connects PC1 to VLAN10
   - Assign the switch port that connects PC2 to VLAN20

- Assign the switch port that connects to the router as a trunk port.

5. Configure the Inter-VLAN routing on R1:
   - Create a subinterface FA 0/0.10 and set the encapsulation to dot1q using VLAN10. Assign the first possible IP address of VLAN10 for this interface.
   - Create a subinterface FA 0/0.20 and set the encapsulation to dot1q using VLAN20. Assign the first possible IP address of VLAN20 for this interface.

6. Configure PC1 with an IP address from VLAN10 and configure PC2 with an IP address from VLAN20. What gateway information will be used for both PCs?

   From PC1, try to ping PC2. The ping should be successful.

7. Configure DHCP on R1. Create a DHCP pool for VLAN10 and VLAN20. Set the correct network and default router information for each pool. Exclude the first 10 IP address from each pool using the command "ip dhcp excluded-address [start IP address] [end IP address]".

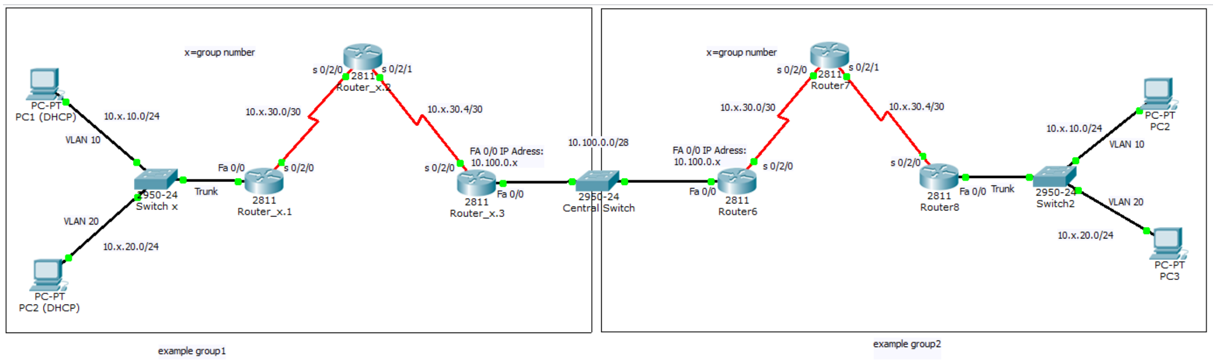
8. Configure the PCs to use DHCP. Check if the PCs received the correct addresses. PC1 should still be able to ping PC2.

9. Configure RIP version2 on all routers. Use the additional commands "version 2" and "no auto-summary". All other commands are the same as in RIP version1.

10. Examine the routing tables of all routers. Which networks are present in the routing table and how are they connected?

11. Ping from the PCs to the FA0/0 interface of Router3. The pings should be successful. Otherwise try to isolate and resolve the problems and examine the routing tables again.

Merging the networks



12. After your group topology is completely functional, merge your topology with all other group topologies into a single network. Disconnect FA0/0 on Router3 from the hub and connect it to the central Switch. The central switch is a single switch that is assigned by the lab administrator and will be used by all groups. Use the central switch port number that reflects your group number.
13. As soon as more than one group is connected to the central switch, examine the changes to the routing tables. The PCs from different groups should now be able to ping each other.
14. Use wireshark to analyze RIPv2. What addresses are used as destination addresses by the routers to exchange routing information? What is the difference to RIPv1?